Online Shopping Behaviour

Predicting regular online shoppers from survey responses (Bangladesh)

1. Summary

The dataset holds 201 survey responses on online shopping habits. After cleaning, 200 responses were used. The task is a binary classification: is a respondent a **regular online shopper** (buys at least once a month) or a **rare/non-shopper**? The data has no explicit target, so this was chosen as the most useful outcome in the survey.

Four models were compared with 5-fold cross-validation and a held-out test set. **Logistic Regression** performed best: ROC-AUC 0.77 and accuracy 74% on the test set (the majority-class baseline scores 64%). Performance is modest, which is expected with about 200 respondents and purely categorical answers. The strongest signals were **participation in promotions** and **confidence in product authenticity**.

2. Data and Preparation

All 17 columns are categorical survey answers. Cleaning steps: removed 1 corrupted row (date-like values in Gender and shopping frequency), merged the "Other" gender response, fixed a typo ("International"), grouped free-text answers in *Frequent_Products*, *Major_Drawback* and *Desired_Improvements* into "Other/Multiple", and one-hot encoded all features.

Target: Regular shopper = 1 (monthly, weekly or multiple times per week; 128 respondents), 0 = rarely or never (72). *Online_Purchase_Proportion* was excluded as it measures nearly the same behaviour as the target (leakage). The data was split 75/25 stratified (150 train, 50 test).

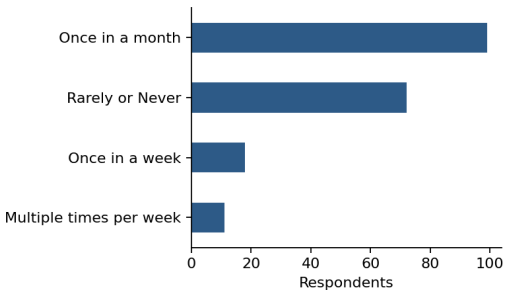


Figure 1. Online shopping frequency

3. Models and Results

Logistic Regression (L2), Random Forest and Gradient Boosting were compared against a majority-class baseline. Class weights were balanced where supported.

Model	CV AUC	CV Acc	Test Acc	Test F1	Test AUC
Baseline (majority)	0.50 ± 0.00	0.64	0.64	0.78	0.50
Logistic Regression	0.71 ± 0.05	0.65	0.74	0.80	0.77
Random Forest	0.68 ± 0.07	0.66	0.72	0.81	0.74
Gradient Boosting	0.62 ± 0.03	0.59	0.66	0.77	0.60

A tuned Random Forest (grid search, depth 3) reached a CV AUC of 0.70, essentially matching Logistic Regression, but with a lower test accuracy (0.64). Gradient Boosting overfit. Logistic Regression was selected as the final model for its accuracy and interpretability.

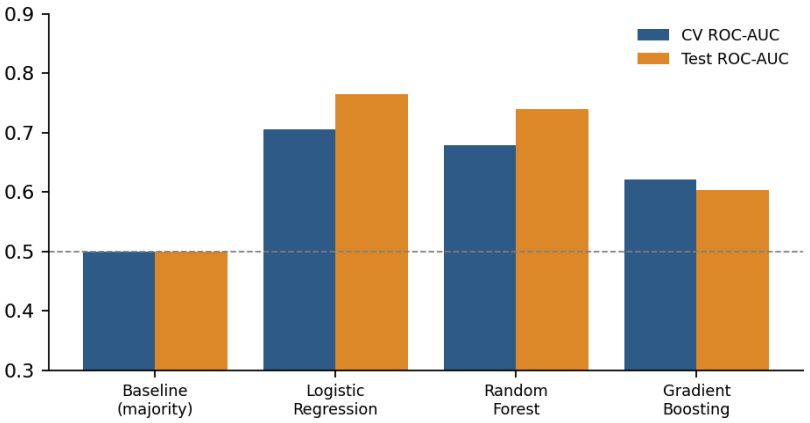


Figure 2. ROC-AUC by model (dashed line = random guessing)

4. Final Model Performance

On the 50 test respondents, the final model correctly classified 37 (74%). It identified 26 of 32 regular shoppers (recall 0.81) and 11 of 18 rare shoppers (recall 0.61).

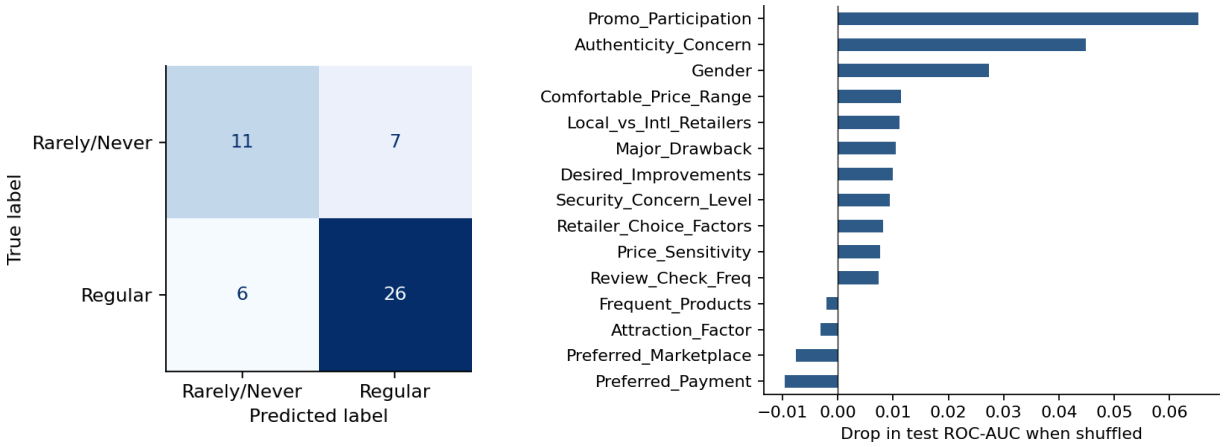


Figure 3. Confusion matrix (left) and permutation feature importance, Random Forest (right)

5. Key Findings

Promotions: 83% of those who frequently participate in promotions shop regularly, versus 45% of those who rarely or never do.

Authenticity trust: only 37% of respondents who are often concerned about authenticity shop regularly, compared with 76% of those mostly confident.

Gender: 71% of female respondents are regular shoppers versus 53% of male respondents.

Other survey answers (payment method, marketplace, security concern, price sensitivity) added little predictive value. Most respondents prefer cash on delivery (83%) and Daraz is the most used marketplace.

6. Limitations and Next Steps

The sample is small (about 200) and the test set only 50, so metrics carry wide uncertainty (CV AUC varies about ±0.05). Responses are self-reported and likely from a convenience sample, so results may not generalise to all of Bangladesh. The importance scores are small and should be read as indications, not proof. Recommended next steps: collect more responses, add age, income and location, and use repeated cross-validation to compare models more reliably.